~~Are Heat-Health Systems Socially Blind? Social Isolation~~Prefecture-Level Older-Adult Solo Household Rate and ~~Dehydration-Related Healthcare~~Large-Volume Infusion Therapy Utilization ~~Across~~in Japan: An Ecological Study

Haruki Saito*[1], Tetsuya Ohira[1,2]

1. Department of Epidemiology, Fukushima Medical University School of Medicine, Fukushima, Japan
2. Radiation Medical Science Center for the Fukushima Health Management Survey, Fukushima Medical University, Fukushima, Japan

*Corresponding author. Email: m211039@fmu.ac.jp

Abstract

Background

~~Heat-health warning systems rely on meteorological indicators including temperature and wet-bulb globe temperature (WBGT), but heat outcomes may also reflect social vulnerability.~~ Older adults living alone ~~face additional risk through limited~~are often considered vulnerable to heat because of reduced informal monitoring, delayed help-seeking, and barriers to cooling. ~~We examined whether prefectural~~Prefecture-level indicators such as elderly solo household rate ~~—~~have been proposed as routinely available markers of this vulnerability, but such indicators may also reflect a ~~community-level marker of social isolation — was associated with dehydration-related~~prefecture's underlying population age structure. We examined the association between elderly solo household rate and large-volume infusion therapy utilization, and tested whether this association remained after accounting for population age structure, healthcare ~~utilization~~supply, and summer heat exposure.

Methods

We conducted an ecological study of 47 Japanese prefectures using fiscal year 2023 data. ~~Infusion~~Large-volume infusion therapy utilization ~~(≥~~(procedure code G004, ≥500 mL) from the National Database (NDB) Open Data was used as ~~a~~the outcome, expressed per 100,000 population~~-level indicator of dehydration-related healthcare utilization.~~. Elderly solo household rate, defined as the percentage of all households consisting of one person aged ≥65 years, was derived from the 2020 National Census~~.~~ ~~Heatwave~~ and used as the primary exposure. We fit cumulative regression models adding, in turn, prefectural ageing rate, general hospital beds per 100,000 population, and official daily-maximum wet-bulb globe temperature (WBGT) days ~~and mean WBGT were calculated from Japan Meteorological Agency data for~~(≥28°C, June–

September 2023~~. Predictors were~~). Nonlinearity, metropolitan influence, and an alternative exposure using the older population as the denominator were also examined ~~using simple linear regression and sensitivity analyses~~.

Results

~~Mean elderly solo household rate was 12.6 ± 1.9% and mean infusion therapy rate was 8,434 ± 2,587 per 100,000 population.~~ Elderly solo household rate was associated with infusion therapy utilization in the unadjusted analysis (β = ~~723.37~~656.2; 95% CI, ~~376.52–1,070.21; R^2 =~~326.9–985.6; p < 0.~~282~~001). Adding prefectural ageing rate attenuated the estimate by 53.3% (β = 306.1; 95% CI, −71.3–683.5; p = 0.~~0001). The~~112), and the association ~~persisted in all sensitivity analyses. Heatwave frequency, WBGT, and air conditioning prevalence were not~~was no longer statistically significant ~~ecological correlates.~~. Further adjustment for healthcare supply and WBGT reduced the estimate toward zero (β = −9.4; 95% CI, −493.9–475.1; p = 0.970).

Conclusions

~~Community-level social isolation was the only ecological factor significantly associated with infusion therapy utilization. Heat-health systems relying solely on meteorological thresholds may overlook socially vulnerable communities. Integrating social vulnerability indicators may improve equitable climate adaptation.~~

The prefectural elderly solo household rate was associated with large-volume infusion therapy utilization in the unadjusted analysis. However, the estimate was substantially attenuated and was no longer statistically significant after adjustment for population age structure. Further adjustment for healthcare supply and summer heat exposure reduced the estimate toward zero. These findings do not support an independent ecological association between living-alone indicators and infusion

therapy utilization and highlight the need to account for population structure when evaluating social vulnerability in heat-health research.

Keywords: heat-health surveillance; ~~social isolation; ~~elderly solo household~~;~~ ~~dehydration~~ rate; population ageing; healthcare utilization; ecological study; Japan

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Introduction

Heat-related illness is an increasingly urgent public health challenge in Japan. In 2023, heatstroke caused 1,580 deaths, and emergency transports for suspected heatstroke reached 91,467 cases (Fire and Disaster Management Agency 2023; MHLW 2024~~)~~). Climate change, population ~~aging~~ageing, and urbanization are expected to increase this burden (IPCC 2021; Günsche et al. 2026~~)~~).

Heat-health responses have therefore emphasized meteorological surveillance. Temperature, humidity, and WBGT are established proximal determinants of heatstroke (Parsons 2006; Blazejczyk et al. 2012~~)~~), and Japan has implemented WBGT forecasting and heat alert systems (Ministry of the Environment Japan n.d~~.)~~.).. These systems are essential, but they primarily describe environmental hazard~~. They~~ and do not ~~necessarily~~by themselves identify ~~communities~~which populations may be least able to receive warnings, act on them, or obtain timely assistance.

~~Heat risk is produced by both environmental exposure and social vulnerability. Older adults are physiologically vulnerable to heat because of~~Older adults living alone are often considered vulnerable to heat because of reduced informal household monitoring (for example, a family member or neighbor noticing signs of heat illness or prompting fluid intake and use of cooling), economic barriers to cooling use, delayed help-seeking, and reduced access to heat-health information, in addition to physiological vulnerability from reduced sweating capacity, diminished thirst perception, impaired thermoregulation, and chronic disease (Kenney and Munce 2003; Bouchama and Knochel 2002~~) Among older adults, those living alone may face additional risk through limited informal monitoring, economic barriers to cooling, delayed help-seeking, and reduced access to heat-health information (~~; Vandentorren et al. 2006; Canouï-Poitrine et al. 2006; MIC 2015; Farbotko and Waitt 2011; Semenza et al. 1996; Naughton et al. 2002; Oberai et al. 2025; Zhang et al. 2025~~)~~). Prefecture-level elderly solo household rate, the percentage of all households consisting of one older adult living alone, has been proposed as a routinely available marker of this social vulnerability. However, because such a measure is calculated over all households, it may also reflect a prefecture's overall population age structure rather than a distinct social characteristic, and this potential confounding has not been formally examined in prior ecological work.

~~Despite this concern, heat-health research and surveillance remain more developed for monitoring meteorological hazard than social vulnerability.~~ Japanese studies have examined individual risk factors and ambulance transport data (Ng et al. 2014; Miyatake et al. 2012~~)~~), but nationwide ecological evidence linking ~~social isolation~~ elderly solo household rate to ~~dehydration-related~~large-volume infusion therapy utilization, and on whether such an association is independent of population age structure, healthcare ~~utilization~~supply, and heat exposure, is limited. We therefore examined whether elderly solo household rate ~~could serve as a practical social vulnerability layer complementing climatic indicators in heat-health surveillance~~was associated with large-volume infusion therapy utilization, and the extent to which this association remained after accounting for prefectural ageing rate, healthcare supply, and official heat-exposure measures.

Materials and methods

1. Study Design

We conducted a nationwide ecological study using Japanese prefectures (N = 47) as the unit of analysis~~.~~; the prefectures ranged in total 2023 population from approximately 0.5 million to 14.1 million, together representing Japan's total population of approximately 124.4 million. Prefecture-level aggregate data were used because individual household and geospatial information ~~is~~are not available from NDB Open Data. The study period was fiscal year 2023 (April 2023 through March 2024); ~~climatic exposures were~~heat exposure was measured ~~at government meteorological stations~~ using official Ministry of the Environment wet-bulb globe temperature (WBGT) monitoring-site observations during the summer season (June–September 2023~~).~~), an unusually hot summer at the time of data collection (Japan Meteorological Agency 2023).

2. Outcome: Large-Volume Infusion Therapy Utilization

Infusion therapy utilization rates were derived from the 10th NDB Open Data. We extracted claims for infusion procedures involving ≥500 mL volume (procedure code~~:~~ ~~G004), commonly used for rehydration in moderate-to-severe dehydration from heat illness or other acute conditions (Bouchama and Knochel 2002) Because the outcome is not heat-specific, we interpreted it as a population-level indicator of dehydration-related healthcare utilization. Rates were calculated per 100,000 population using the 2020 National Census.~~ G004), used for volume repletion in a range of clinical contexts including, but not limited to, heat-related dehydration; the code is not diagnosis-specific. We describe the outcome as large-volume infusion therapy utilization, a non-specific healthcare-utilization measure that may include a heat-related dehydration component. Rates were calculated per 100,000 population using the 2023 population estimate, temporally aligned with the FY2023 NDB claims period (a sensitivity analysis using the 2020 Census population denominator is reported in Online Resource 1). We independently reviewed the complete public file structure of the 10th NDB Open Data and confirmed that prefecture-by-age cross-tabulated counts for G004 are not publicly available; sex/age and prefecture counts exist only as separate marginal tables. We therefore could not construct an age-specific or age-standardized outcome and instead adjusted for prefectural ageing rate directly (Section 5). As a construct validity check, we examined the cross-prefectural correlation between infusion therapy utilization ~~rate~~ and the summer 2023 heatstroke ambulance transport rate ~~derived~~ from the Fire and Disaster Management Agency (FDMA) daily transport report (~~Online Resource 1, panel a) (~~Fire and Disaster Management Agency 2023~~)~~); this provides limited support for the presence of a heat-related component in the outcome rather than validating it as a direct heat-illness measure.

3. Exposure: Elderly Solo Household Rate

Elderly solo household rate was defined as the percentage of all households consisting of one person aged ≥65 years, derived from the 2020 National Census. ~~This measure was interpreted as a community-level marker of social isolation.~~We interpret this measure as a prefecture-level indicator of the burden of older adults living alone relative to all households, rather than a direct individual-level measure of social isolation. As a complementary sensitivity analysis, we also constructed an alternative exposure using the prefecture's population aged ≥65 years as the denominator (percentage of the 65+ population living alone; Section 9; Online Resource 1).

4. Climatic Covariates

~~Heatwave exposure was defined as the number of days with maximum temperature ≥35°C during June–September 2023, averaged across Japan Meteorological Agency (JMA) weather stations within each prefecture. Mean WBGT was estimated from daily mean temperature (T, °C) and relative humidity (RH, %) using a simplified approximation: wet-bulb temperature $T_w = T - (100 - RH)/5$; $WBGT = 0.7 \times T_w + 0.3 \times T$, averaged over all JMA stations and the summer season (June–September 2023) (Patel et al. 2013; Liljegren et al. 2008) This formula provides an indoor-equivalent WBGT approximation from routinely available meteorological observations. To assess whether additional dimensions of temperature exposure might explain the infusion therapy pattern, four supplementary climate variables were derived from the same JMA station data: (1) mean daily maximum temperature (°C); (2) standard deviation of daily maximum temperature as a measure of temperature variability; (3) mean diurnal temperature range (daily maximum minus daily minimum temperature, °C); and (4) rapid-change days, defined as the number of days on which the day-to-day change in daily maximum temperature exceeded 5°C. These four variables were examined in univariate regression against infusion therapy utilization and results are presented in Table 2.~~

~~5. Additional Covariates~~

Heat exposure was assessed using official wet-bulb globe temperature (WBGT) observations from the Ministry of the Environment's Heat Illness Prevention Information monitoring network for June–September 2023. For each monitoring site, we counted the number of days on which daily maximum WBGT met or exceeded each threshold, and then averaged these site-specific day-counts within each prefecture (mean 17.9 sites per prefecture; range 5–163). The primary heat metric used in the revised analysis was the prefecture-mean number of days with daily maximum WBGT ≥28°C; days with WBGT ≥31°C, days with WBGT ≥33°C, and cumulative WBGT excess above 28°C were examined as sensitivity heat metrics (Section 6; Online Resource 1). For descriptive purposes (Table 1), we also report the number of days with daily maximum temperature ≥35°C, averaged across Japan Meteorological Agency stations within each prefecture; this variable was not used as an adjustment variable in the primary models.

5. Prefectural Ageing Rate and Healthcare Supply

Prefectural ageing rate, defined as the percentage of the population aged ≥65 years and calculated from the Statistics Bureau of Japan's population estimate as of 1 October 2023, was included as the primary demographic adjustment variable, temporally aligned with the outcome denominator. General hospital beds per 100,000 population (2023 Survey of Medical Institutions, Ministry of Health, Labour and Welfare) were included as the primary healthcare-supply adjustment variable, with general clinics per 100,000 population (same source and year) examined as a sensitivity analysis substituted for beds. Air conditioning prevalence (proportion of households owning at least one air conditioning unit, ~~%) was~~%), obtained from the 2014 National Survey of Family Income and Expenditure~~. Aging rate (proportion of individuals aged ≥65 years) was not available in the analysis dataset and~~ , is retained for descriptive purposes (Table 1) but was ~~therefore~~ not included in the primary adjusted models, because these data describe ownership rather than use.

6. Statistical Analysis

~~We summarized all variables using means, standard deviations, medians, and ranges. Given substantial collinearity among ecological predictors (variance inflation factor >10 in preliminary multivariable models), each factor was examined separately using simple linear regression with infusion therapy utilization as the outcome. This~~

~~approach estimated policy-interpretable population-level associations rather than a predictive multivariable model.~~

~~Three sensitivity analyses assessed the robustness of the primary finding: (1) **Outlier exclusion**: We calculated Cook's distance for each prefecture and excluded observations exceeding 4/N (N = 47 = 0.085). (2) **Tokyo exclusion**: We repeated the base model excluding Tokyo, given its extreme urbanization and population density. (3) **Stratified analysis**: We stratified prefectures by the median elderly solo household rate (12.3%) and fit separate regression models within each stratum to explore effect heterogeneity across the exposure distribution.~~

All models used the originally specified elderly solo household rate as the exposure. We fit four cumulative models: the exposure alone (Model O-A); Model O-A plus prefectural ageing rate (Model O-B); Model O-B plus general hospital beds per 100,000 (Model O-C); and Model O-C plus days with WBGT ≥28°C (Model O-D). We did not use multicollinearity as a reason to avoid adjustment; variance inflation factors for all covariates in all models were below 3.1 (Table 2). For every model we report the unstandardized coefficient with heteroskedasticity-consistent (HC3) robust 95% confidence intervals and p-value, the fully standardized coefficient, adjusted R^2, the partial R^2 for the exposure (based on the reduction in ordinary least-squares residual sum of squares, independent of the standard-error estimator), and the corrected Akaike information criterion (AICc).

We conducted the following sensitivity analyses on the fully adjusted model: substituting general clinics for hospital beds; excluding Tokyo, Osaka, and Kanagawa jointly; leave-one-prefecture-out re-estimation (47 iterations); and separately substituting the primary heat metric with WBGT ≥31°C days, WBGT ≥33°C days, and cumulative WBGT excess, applying Benjamini-Hochberg false-discovery-rate correction across this exploratory heat-metric set. As a further sensitivity analysis, we repeated all models using the 2020 Census population denominator for the outcome (Online Resource 1). We assessed nonlinearity by comparing the linear model with a centered quadratic exposure term and a natural cubic spline (3 degrees of freedom), using partial F-tests and AICc; a median-split stratified analysis is reported in Online Resource 1 and did not support a simple dose-response interpretation. We also examined an alternative exposure using the older population as the denominator (Section 9).

Regression coefficients (β), 95% confidence intervals (CIs), R^2, adjusted R^2, and p-values are reported~~.~~, together with regression diagnostics (residuals, leverage, Cook's distance, DFBETAs) for the fully adjusted model. Statistical significance was defined as two-sided $p < 0.05$~~.~~, interpreted alongside effect estimates and confidence intervals rather than significance alone. Analyses were performed using Python (version 3.14), statsmodels (version 0.14), scikit-learn, and matplotlib.

This study used publicly available aggregate data; individual informed consent was not required, and institutional ethics review was not applicable in accordance with Japanese ethical guidelines for epidemiological research.

Results

1. Descriptive Statistics

Across the 47 Japanese prefectures, ~~the~~ mean elderly solo household rate was 12.6 ± 1.9% (range~~:~~ 9.4–17.8~~%)~~ (%; Table 1). Mean large-volume infusion therapy utilization was 8,~~434~~022 ± 2,~~587~~389 per 100,000 population (range~~:~~ 4,~~267~~231–13,~~697~~234). Mean ~~heatwave~~ prefectural ageing rate was 31.6 ± 3.3% (range 22.8–39.0%). Mean general hospital beds were 794 ± 149 per 100,000 (range 509–1,147). Mean number of days

~~(~~with daily maximum WBGT ≥28°C (official Ministry of the Environment data) was 72.9 ± 14.1 (range 27.3–112.3); for descriptive comparison, mean number of days with maximum temperature ≥35°C~~) were~~ was 17.9 ± 12.1 ~~days~~ (range~~:~~ 0.0–44.0~~ days), and mean summer WBGT was 23.2 ± 1.0°C (range: 19.8–26.3°C).~~). Mean air conditioning prevalence was 89.7 ± 13.6% (range~~:~~ 26.6–98.2%).

~~2. Univariate Regression Analysis~~

2. Association Between Elderly Solo Household Rate and Infusion Therapy Utilization: Unadjusted and Adjusted Models

Elderly solo household rate was ~~the only significant predictor of~~ associated with infusion therapy utilization in ~~univariate regression (~~the unadjusted model (Model O-A: β = ~~723.37~~656.2; 95% CI, ~~376.52–1,070.21~~326.9–985.6; standardized β = 0.521; R^2 = 0.~~282; *p* = 0.0001) (~~272; adjusted R^2 = 0.256; p < 0.001; Table 2~~). Each~~, Fig. 1). Adding prefectural ageing rate (Model O-B) changed the coefficient to 306.1 (95% CI, −71.3–683.5; standardized β = 0.243; p = 0.112), a descriptive attenuation of 53.3%; the 95% confidence interval crossed zero and statistical significance was lost, although the interval remains compatible with both a modest positive association and little or no association. Adding general hospital beds per 100,000 (Model O-C) reduced the coefficient further (β = 94.1; 95% CI, −302.4–490.6; p = 0.642); the estimate was similarly attenuated when clinics were substituted for beds (Online Resource 1). Adding WBGT ≥28°C days (Model O-D) reduced the coefficient to near zero and reversed its sign (β = −9.4; 95% CI, −493.9–475.1; p = 0.970). Maximum VIF across all models was 3.01 (Table 2).

3. Influence Diagnostics and Sensitivity Analyses

Cook's distance for the fully adjusted model identified Hokkaido as the most influential prefecture (Cook's D = 0.667, exceeding the 4/N = 0.085 threshold by approximately 7.8-fold), followed by Kochi (0.190) and Hiroshima (0.144); Hokkaido's WBGT summary averages 163 monitoring sites spanning substantial internal climatic heterogeneity. Leave-one-~~percentage-point increase~~prefecture-out re-estimation of the fully adjusted model (47 iterations) produced coefficients ranging from −107.9 to 92.8, with the 95% CI crossing zero in every iteration, indicating that the attenuated result was not driven by any single prefecture, including Hokkaido. Excluding Tokyo, Osaka, and Kanagawa jointly did not materially change the result ($\beta = -42.0$; 95% CI, −645.6–561.6; $p = 0.891$).

4. Nonlinearity

A quadratic term for elderly solo household rate ~~was associated with approximately 723 additional infusion procedures per 100,000~~ did not improve model fit relative to the linear fully adjusted model (partial F test $p = 0.142$; AICc 845.6 versus 845.9), nor did a natural cubic spline (3 degrees of freedom; $p = 0.141$; AICc 845.9). A median-split stratified analysis, in which the association was present in the lower stratum but absent in the higher stratum, is reported in Online Resource 1; it did not support a simple dose-response interpretation and is not used as a primary result.

5. Alternative Exposure (Denominator) Analysis

As a complementary sensitivity analysis, we examined an alternative exposure defined as the percentage of the prefecture's population~~. Heatwave days, WBGT, and air conditioning prevalence were not significantly associated with infusion therapy~~ aged ≥65 years who live alone (denominator: 65+ population, rather than all households). This alternative measure showed no association with infusion utilization~~. Four additional climate variables — mean daily maximum temperature, temperature variability (SD of daily maximum temperature), mean diurnal temperature range, and rapid-change days (≥5°C day-to-day change) — were also not significantly associated (all $p > 0.34$) (Table 2~~, either unadjusted (standardized β = 0.0003; 95% CI, −0.327–0.328; p = 0.998) or in the analogous fully adjusted model (standardized β = −0.034; 95% CI, −0.298–0.231; p = 0.803), and this pattern was unchanged across the sensitivity analyses described above (Online Resource 1).

~~3. Sensitivity Analysis: Outlier Exclusion~~

~~Cook's distance analysis identified Hokkaido and Kochi as influential observations exceeding the threshold of 4/N = 0.085. After excluding these two prefectures (N = 45), the association strengthened (β = 925.11; 95% CI, 564.74–1,285.47; $R^2 = 0.384$; $p < 0.001$) (Table 2, Panel B).~~

~~4. Sensitivity Analysis: Tokyo Exclusion~~

~~After excluding Tokyo (N = 46), the association remained statistically significant (β = 696.92; 95% CI, 352.45–1,041.39; $R^2 = 0.274$; $p = 0.0002$) (Table 2, Panel B), indicating that the finding was not driven by the unique characteristics of Japan's largest metropolitan area.~~

~~5. Stratified Analysis by Elderly Solo Household Rate~~

~~When prefectures were stratified by the median elderly solo household rate (12.3%), the association differed between strata. In the lower stratum (N = 23, rate <12.3%), the association was statistically significant (β = 1,254.30; 95% CI, 7.69–2,500.92; $R^2 = 0.173$; $p = 0.049$). In the higher stratum (N = 24, rate ≥12.3%), the association was not statistically significant (β = 264.50; 95% CI, −461.52–990.53; $R^2 = 0.025$; $p = 0.458$)~~

~~(Table 2, Panel B). The estimated coefficient differed between strata (lower stratum β = 1,254.30 vs. higher stratum β = 264.50).~~

6. ~~Construct Validity:~~Comparison with FDMA Heatstroke Ambulance Transport~~ Validation~~

As a construct validity check, infusion therapy utilization ~~rate~~ was examined in relation to the summer 2023 heatstroke ambulance transport rate ~~(per 100,000 population) derived~~ from FDMA (Fire and Disaster Management Agency 2023~~)~~). A ~~significant~~ positive association was observed (β = 70.19; 95% CI, 36.85–103.53; r = 0.534; p < 0.001; Online Resource 1~~, panel a), supporting~~), which provides limited support for the ~~use~~presence of a heat-related component in the infusion therapy outcome, rather than validating it as a ~~population-level indicator~~direct measure of heat-related dehydration.

~~7. Negative Control: Outpatient Utilization Rate~~

~~To assess whether the primary association might reflect general healthcare access rather than heat-related dehydration specifically, we examined the association between elderly solo household rate and general outpatient utilization rate derived from the R5 Patient Survey (Ministry of Health, Labour and Welfare), which represents all outpatient visits per 100,000 population. General outpatient utilization was significantly associated with elderly solo household rate (β = 123.52; 95% CI, 44.30–202.74; R^2 = 0.180; p = 0.003; Online Resource 1, panel b). However, the association was substantially weaker than that observed for infusion therapy: the β coefficient was approximately six-fold smaller (123.52 vs. 723.37) and explained less variance (R^2 = 0.180 vs. 0.282). This comparison suggests that while elderly solo household rate captures some component of general healthcare utilization, the much stronger~~

~~association with infusion therapy may reflect heat-related dehydration beyond what would be expected from general healthcare access patterns alone.~~

7. Comparison-Outcome Analysis

A comparison with general outpatient utilization (R5 Patient Survey) did not support outcome specificity for infusion therapy and is reported in the supplementary material (Online Resource 1), using standardized coefficients only; the previously reported raw-coefficient comparison is not valid because the two outcomes are measured on different scales and has been removed.

Discussion

~~Heat-health surveillance has traditionally focused on *where* heat occurs. Our findings suggest that surveillance should also identify *where* people are least able to respond to heat. In this~~ This nationwide ecological analysis~~,~~ examined whether elderly solo household rate, a routinely available prefecture-level indicator, was ~~robustly~~ associated with large-volume infusion therapy utilization~~, while heatwave days,~~ and ~~WBGT were not significant ecological correlates. This pattern suggests that social isolation may capture a surveillance-relevant dimension of heat vulnerability not represented by meteorological hazard alone.~~

~~The~~whether any such association ~~is plausible because living alone can amplify heat risk between~~ was independent of population age structure, healthcare supply, and summer heat exposure ~~and care. Older adults may have~~. An association was present in the unadjusted analysis but was substantially attenuated (53.3%) and lost statistical significance after adjustment for prefectural ageing rate alone, and was further attenuated toward zero after additional adjustment for healthcare supply and heat exposure. An alternative exposure using the older population as the denominator showed no association with the outcome at any stage. These findings do not support an independent ecological association between elderly solo household rate and infusion therapy utilization.

Physiological vulnerability to heat among older adults, including reduced thermoregulation and blunted thirst perception, remains biologically plausible and is well documented (Kenney and Munce 2003; Bouchama and Knochel 2002; Bach et al. 2024; Blatteis 2012~~) while living alone may reduce informal prompts to~~). However, the present ecological results indicate that a prefecture-level living-alone indicator does not, by itself, capture an independent surveillance-relevant signal once population age structure is taken into account.

Regional infusion use ~~air conditioning, drink fluids, or seek care. Economic concerns can further limit cooling use despite high air conditioning ownership (MIC 2015; Farbotko and Waitt 2011) Social isolation may also delay recognition of confusion, dehydration, or reduced activity, and may limit receipt of digital heat alerts or interpretation of~~may also reflect healthcare supply and local treatment practice in addition to underlying morbidity: the estimate was further attenuated after adjusting for general ~~heat-health advice (Semenza et al. 1996; Naughton et al. 2002; Oberai et al. 2025; Zhang et al. 2025) (Berkman et al. 2011)~~hospital beds, and this pattern was unchanged when clinics were substituted for beds. We did not test whether supply-induced demand specifically explains infusion use; we show only that accounting for healthcare supply further attenuates the association between elderly solo household rate and infusion therapy utilization.

Our findings ~~align with~~ do not contradict international evidence that social isolation ~~increases~~is associated with heat-related mortality at the individual and small-area level, including during the 1995 Chicago and 2003 European heat waves

(Vandentorren et al. 2006; Canouï-Poitrine et al. 2006; Semenza et al. 1996; Conti et al. ~~2005) This study extends that literature by using nationwide administrative and census data to link an available household-composition indicator with dehydration-related healthcare utilization~~2005). Ecological associations and individual-level associations address different questions, and our attenuated ecological finding at the prefecture level should not be read as evidence against individual-level effects of social isolation on heat vulnerability, which this design cannot test.

~~Japanese studies have identified advanced age and comorbidity as heatstroke risk factors (Ng et al. 2014; Miyatake et al. 2012) but fewer have examined social isolation as~~ The absence of a ~~prefecture-level surveillance indicator. The estimate of approximately 723 additional infusion procedures per 100,000 population~~robust independent association for ~~each one-percentage-point increase in~~ elderly solo household rate ~~may be useful for prioritizing outreach and resource allocation.~~

~~The absence of significant associations for heatwave days and WBGT~~ should not be interpreted as evidence that ~~climate~~heat exposure is unimportant. ~~Infusion therapy is~~Alternative WBGT metrics did not ~~heat-specific, prefecture~~materially alter the interpretation; detailed exploratory results are reported in Online Resource 1. Prefecture-level climatic averages may also miss neighborhood and indoor exposures, and regional heat adaptation may attenuate simple ecological climate-health associations (Hondula et al. 2017; Harlan et al. 2013; Oka et al. ~~2023) These considerations support combining meteorological indicators with social and built-environment indicators.~~2023).

These findings ~~have implications for heat-health surveillance and climate adaptation policy. WBGT-based alerts remain essential, but they cannot determine who is likely to be missed. Prefectures with high elderly solo household rates could prioritize~~

~~welfare checks, community-based *Mimamori* programs, cooling support, and accessible, plain-language health communication during heat alerts.~~

~~A layered system could combine meteorological hazard forecasts with social~~ suggest that prefecture-level social-vulnerability indicators such as elderly solo household rate ~~and cooling access (Romitti et al. 2025) Such an approach would not replace climate alerts; it would guide where outreach, social care, and clinical preparedness~~ should ~~be intensified. Heat adaptation has primarily been framed~~not be interpreted as ~~a meteorological problem; our findings suggest it should also be understood as a social care problem, supporting integration of social~~ independent heat-vulnerability ~~into existing heat-health surveillance~~markers without accounting for population age structure, healthcare supply, and improved heat-exposure measures. This is consistent with, rather than ~~replacement of meteorological warning systems~~a departure from, the original rationale for examining social indicators in heat-health surveillance: it indicates that such indicators require careful demographic validation before being proposed for that purpose.

Strengths include nationwide coverage of all 47 prefectures, use of NDB Open Data ~~with~~ , direct re-examination of the originally specified exposure and outcome, replacement of the original simplified WBGT approximation with official daily-maximum WBGT observations, and a comprehensive ~~insurance claims, and robustness across multiple~~set of sensitivity, influence, and nonlinearity analyses. ~~The ecological design also demonstrates how routinely available social indicators may strengthen heat-health surveillance.~~

Several limitations should be considered. First, this is an ecological design; associations at the prefecture level may not reflect individual-level relationships~~. We~~, and we cannot determine whether individuals living alone within a prefecture had higher infusion therapy utilization than those living with family. ~~Second, this study did~~

~~not directly evaluate heat warning systems; “socially blind” should be interpreted as a policy hypothesis, not a direct criticism of existing systems. Third, the cross-sectional design precludes causal inference~~Second, elderly solo household rate is a household-composition measure and is not equivalent to social isolation. Third, infusion therapy utilization (G004) is not specific to dehydration or heat illness and may be used for volume repletion in other contexts, including perioperative fluid management, gastrointestinal illness, infections, and general supportive care; monthly aggregation of claims is not available in prefecture-level NDB Open Data, so a summer peak in utilization could not be verified directly.

~~Fourth, because multivariable ecological modeling was not feasible owing to substantial collinearity, residual confounding cannot be excluded. Fifth, infusion therapy utilization is not heat-specific and captures dehydration from multiple causes, which may bias associations toward the null. Heat-specific diagnoses would strengthen inference but are not available in prefecture-level NDB Open Data. Additionally, monthly aggregation of procedure claims is not available in prefecture-level NDB Open Data; therefore, it was not possible to verify that infusion therapy utilization peaks during summer months, as would be expected if the outcome primarily captures heat-related dehydration. As a negative control, general outpatient utilization rate (R5 Patient Survey) was also significantly associated with elderly solo household rate (β = 123.52; 95% CI, 44.30–202.74; R^2 = 0.180; p = 0.003; Online Resource 1, panel b); however, the β coefficient was approximately six-fold smaller than for infusion therapy (123.52 vs. 723.37) and explained less variance (R^2 = 0.180 vs. 0.282), suggesting partial specificity of the primary association beyond general healthcare access. Cross-validation against the 2023 summer FDMA heatstroke ambulance transport rate (r = 0.53, p < 0.001; Online Resource 1, panel a) further supported the use of infusion therapy as a population-level proxy for heat-related dehydration.~~

~~Sixth, unmeasured confounding by socioeconomic status, healthcare access, chronic disease, medication use, housing, and urban heat islands may remain. Climatic measures were prefecture-level averages and did not capture indoor or neighborhood exposures.~~

~~Seventh, WBGT was estimated using a simplified approximation (Tw = T − (100 − RH)/5; WBGT = 0.7 × Tw + 0.3 × T) applied to daily mean temperatures. This indoor-equivalent formula may underestimate peak outdoor heat stress, which could attenuate the observed association between WBGT and healthcare utilization.~~ Fourth, prefecture-by-age cross-tabulated G004 counts are not publicly available in the 10th NDB Open Data, precluding an age-specific or age-standardized outcome; we addressed this by adjusting for prefectural ageing rate directly. Fifth, the analysis covers a single fiscal year during an unusually hot summer (2023; Japan Meteorological Agency 2023), limiting generalizability to other years. Sixth, with N = 47, adjusted confidence intervals remain wide, and this analysis cannot precisely distinguish a small residual association from no association. Seventh, unmeasured confounding cannot be excluded, including reduced thirst perception and lower fluid intake among older adults, chronic disease prevalence, medication use, infections, gastrointestinal illness, and other individual clinical factors associated with infusion therapy that are not captured at the prefecture level.

Eighth, climatic measures are prefecture-level averages and do not capture indoor or neighborhood exposures; the WBGT summary for Hokkaido in particular averages 163 monitoring sites across substantial internal climatic heterogeneity and was the most influential single observation in regression diagnostics, although the qualitative conclusion was unchanged when Hokkaido was excluded (leave-one-prefecture-out analysis). Ninth, source years differ across variables (2020 Census for elderly solo household rate; 2023 estimates for population, ageing rate, and outcome denominator; 2023 Survey of Medical Institutions for healthcare supply; 2014 survey for air conditioning prevalence), and we aligned the outcome denominator to 2023 to match the NDB claims period while retaining a 2020 Census-based denominator analysis as a sensitivity check (Online Resource 1).

Finally, air conditioning prevalence ~~was derived from 2014 survey data; while national ownership rates have remained high in Japan, the extreme lower range observed in this study (26.6% in one prefecture) likely~~(2014 survey) reflects ~~historical underuse in cold-climate prefectures that may have changed in recent years.~~ ownership, not actual use, and cannot establish whether cooling was applied; it is reported descriptively (Table 1) and was not included in the primary adjusted models. The present findings should be interpreted as evidence for cautious, demographically informed use of social-vulnerability indicators in heat-health surveillance ~~prioritization,~~ rather than ~~causal~~as evidence ~~of~~ against considering social factors altogether.

Given the ecological, cross-sectional, and single-year design, these findings should be considered hypothesis-generating. These considerations indicate that firmer conclusions about an independent heat-~~related illness~~vulnerability effect of household living arrangement would require age-specific outcome data, multi-year designs, and finer-grained exposure measurement, in addition to the demographic caution highlighted above.

Conclusions

In this nationwide ecological study, elderly solo household rate was associated with ~~dehydration-related healthcare~~ large-volume infusion therapy utilization ~~across Japanese prefectures~~ in ~~univariate analyses. The~~unadjusted analysis. However, this association was ~~consistent across multiple sensitivity analyses, whereas heatwave days~~substantially attenuated and ~~WBGT were not~~no longer statistically significant ~~ecological correlates~~after adjustment for prefectural ageing rate, and was further attenuated toward zero after additional adjustment for healthcare supply and summer heat exposure.

~~These findings suggest that social isolation may be a missing layer in heat-health surveillance. Meteorological thresholds identify hazardous heat, but they may not identify communities where older adults are least able to receive warnings, initiate protective behavior, or obtain assistance. Integrating routinely available social indicators into existing surveillance systems may improve equitable heat adaptation in aging societies.~~

These findings do not support elderly solo household rate as an independent ecological correlate of infusion therapy utilization once population age structure, healthcare supply, and heat exposure are taken into account. Given the ecological, cross-sectional, and single-year design, these findings should be regarded as hypothesis-generating. They highlight the need for careful demographic validation, including explicit adjustment for age structure and attention to denominator choice, before routinely available social-vulnerability indicators are proposed for heat-health surveillance.

Future individual-level and longitudinal studies, and age-specific administrative data where available, are needed to ~~confirm these population-level findings and to~~ clarify ~~pathways linking social isolation,~~whether living alone carries an independent heat-vulnerability signal beyond population age structure, and to establish the role of healthcare supply and improved heat ~~exposure, and dehydration-related healthcare use~~ exposure measures in explaining prefecture-level utilization patterns.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability

The NDB Open Data used in this analysis are publicly available from the Ministry of Health, Labour and Welfare of Japan (https://www.mhlw.go.jp/stf/seisakunitsuite/bunya/0000177182.html). The analysis code ~~is~~, derived data, and provenance records are openly available on GitHub (https://github.com/haruki00430/NDB~~_XXX_heatwave_heatstroke~~-solo-household-infusion-therapy-japan) and archived on Zenodo (https://doi.org/10.5281/zenodo.20740375~~).~~), which always resolves to the most recent version.

Author Contributions

Haruki Saito: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review and editing.

Tetsuya Ohira: Conceptualization, Supervision, Writing – review and editing.

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Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation ~~and writing~~ of ~~this work~~the original submission, the authors used AI-assisted tools to support manuscript drafting and statistical analysis scripting. Cursor 3.0 (Anysphere) and Google Antigravity (Google) were used for AI-assisted writing and Python code development. Large language models used through these platforms included Claude Sonnet 4.6 and Claude Opus 4.8 (Anthropic~~) and~~), GPT-5.5 (OpenAI~~)~~), and Gemini 3 Pro (Google). ~~These tools were used only for text drafting and code generation; no generative AI or~~ During preparation of the revision, the authors used AI-assisted tools ~~were used~~(Claude Sonnet 5, Anthropic, accessed via Claude Code) to ~~create, alter, or otherwise process any figures, images, or artwork in this manuscript.~~support text drafting and the development of data-processing and statistical-analysis code. The authors executed the code, verified all source data, numerical results, tables, and figures, and independently determined the statistical methods and interpretation. No generative image tools were used. The authors reviewed and edited all ~~AI-assisted~~ outputs and ~~were responsible for the study design, selection of statistical methods, interpretation of findings, conclusions, and final reference list. The authors~~ take full responsibility for the ~~integrity and accuracy of the~~ final ~~content~~manuscript. AI was not listed as an author.

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Tables and Figures

Table 1. Descriptive Statistics of Prefecture-Level Variables (N = 47)

Variable	Unit	N	Mean	SD	Min	Median	Max
Outcome variable							
~~Infusion~~Large-volume infusion therapy ~~rate~~ ~~(≥~~utilization (G004, ≥500 mL)	per 100,000	47	8,~~434~~022.2	2,~~587~~389.3	4,~~267~~231.3	7,~~962~~608.7	13,~~697~~234.1
Exposure variable							
Elderly solo household rate	%	47	12.6	1.9	9.4	12.3	17.8
Alternative exposure (sensitivity) variable							
% of population aged ≥65 living alone	%	47	17.7	3.1	12.1	17.1	26.1

Variable	Unit	N	Mean	SD	Min	Median	Max
Adjustment variables							
Prefectural ageing rate	%	47	31.6	3.3	22.8	31.7	39.0
General hospital beds	per 100,000	47	794.2	148.5	508.7	800.3	1,146.5
General clinics (sensitivity)	per 100,000	47	84.5	11.9	61.8	85.9	113.0
Climatic variables							
Days with WBGT ≥28°C (official, primary)	days	47	72.9	14.1	27.3	74.7	112.3
Days with WBGT ≥31°C (sensitivity)	days	47	28.7	10.6	6.0	26.5	61.6
Days with WBGT ≥33°C (sensitivity)	days	47	2.9	2.7	0.0	1.7	10.8
~~Heatwave~~ Days with maximum temperature ≥35°C (descriptive only)~~days (≥35°C)~~	days	47	17.9	12.1	0.0	17.0	44.0
~~Mean WBGT (June–Sep)~~	~~°C~~	~~47~~	~~23.2~~	~~1.0~~	~~19.8~~	~~23.4~~	~~26.3~~
Covariate ~~variable~~(descriptive only)							
Air conditioning prevalence	%	47	89.7	13.6	26.6	94.8	98.2

WBGT, wet-bulb globe temperature; SD, standard deviation. WBGT days are based on official Ministry of the Environment daily-maximum observations (June–September 2023). Days with maximum temperature ≥35°C are shown for descriptive comparison and were not used as an adjustment variable. Air conditioning prevalence ~~from~~ (2014 National Survey of Family Income and Expenditure) reflects ownership, not use, and was not used as an adjustment variable. Ageing rate and healthcare supply are the primary demographic and healthcare-supply adjustment variables in this analysis.

Table 2. Hierarchical Regression ~~Analyses: Associations with~~Models: Elderly Solo Household Rate and Large-Volume Infusion Therapy Utilization

~~Panel A. Univariate Linear Regression — All Predictors~~ (N = 47)

~~Variable~~Model	Cumulative adjustment	N	Unstd. β (HC3 95% CI)		Std. β (HC3 95% CI)	p	Exposure partial R^2	~~Adj.~~Adjusted R^2	
~~Elderly solo household rate (per 1%)~~O-A	~~723.37 (376.52, 1,070.21)~~Unadjusted	47	656.2 (326.9, 985.6)		~~0.0001~~0.521 (0.260, 0.783)	≤0.~~282~~001	0.~~266~~272	0.256	
~~Air conditioning prevalence (per 1%)~~O-B	~~31.23 (−24.98, 87.45)~~+ ageing rate	47	306.1 (-71.3, 683.5)		~~0.269~~0.243 (-0.057, 0.543)	0.~~027~~112	0.~~005~~056	0.365	847.1
O-C	+ ageing rate + hospital beds	47	~~Heatwave~~ 94.~~days (per~~ 1 ~~day~~(-302.4, 490.6)	~~−33.96 (−97.42, 29.51)~~	~~0.287~~0.075 (-0.240, 0.390)	0.~~025~~642	0.~~003~~004	0.386	
O-D	~~Mean WBGT (per 1°C)~~+ ageing	~~94.3~~	-9.4 (-493.9, 475.1)		~~0.802~~-0.007 (-0.392, 0.377)	0.970	≤0.001	~~−~~0.~~01~~422	

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rate + hospital beds + WBGT≥28d	~~2 (−658.68,847.32)~~ 47			
~~*Additional climate variables*~~				
~~Mean daily maximum temperature (per 1°C)~~	~~−145.83 (−861.66, 570.00)~~	~~0.684~~	~~0.004~~	~~−0.018~~
~~SD of daily maximum temperature (per 1°C)~~	~~−586.99 (−2,060.21, 886.23)~~	~~0.427~~	~~0.014~~	~~−0.008~~
~~Mean diurnal temperature range (per 1°C)~~	~~−307.32 (−961.69, 347.06)~~	~~0.349~~	~~0.019~~	~~−0.003~~
~~Rapid-change days ≥5°C (per 1 day)~~	~~−42.32 (−190.74, 106.10)~~	~~0.569~~	~~0.007~~	~~−0.015~~

~~**Panel B. Sensitivity Analyses — Elderly Solo Household Rate → Infusion Therapy Utilization**~~

~~**Analysis**~~	~~**N**~~	~~**β (95% CI)**~~	~~***p***~~	~~**R^2**~~	~~**Notes**~~
~~**Base model**~~	~~47~~	~~**723.37 (376.52, 1,070.21)**~~	~~**0.0001**~~	~~**0.282**~~	~~–~~
~~**Outlier exclusion**~~	~~45~~	~~**925.11 (564.74, 1,285.47)**~~	~~**<0.001**~~	~~**0.384**~~	~~Hokkaido, Kochi excluded~~
~~**Tokyo exclusion**~~	~~46~~	~~**696.92 (352.45, 1,041.39)**~~	~~**0.0002**~~	~~**0.274**~~	~~Tokyo excluded~~
~~**Stratified: High elderly solo HH rate (≥12.3%)**~~	~~24~~	~~264.50 (−461.52, 990.53)~~	~~0.458~~	~~0.025~~	~~–~~
~~**Stratified: Low elderly solo HH rate (<12.3%)**~~	~~23~~	~~**1,254.30 (7.69, 2,500.92)**~~	~~**0.049**~~	~~**0.173**~~	~~–~~

~~**Bold** indicates $p < 0.05$. Panel A: additional climate variables derived from Japan Meteorological Agency daily temperature data (June–September 2023). Panel B:~~

~~outlier detection used Cook's distance threshold 4/N = 0.085; Hokkaido and Kochi exceeded this threshold. HH, household.~~

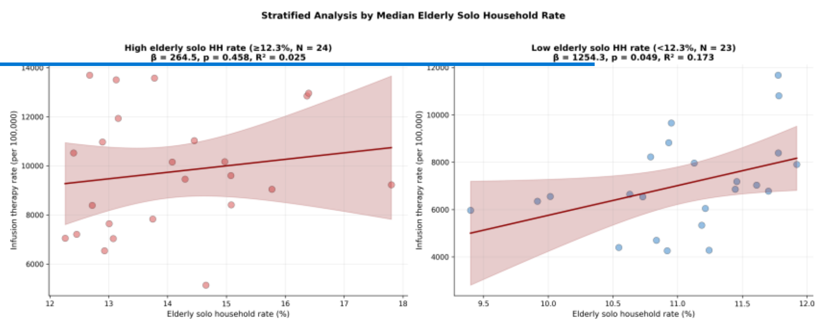
Model O-A: exposure alone. Model O-B: Model O-A + prefectural ageing rate. Model O-C: Model O-B + general hospital beds per 100,000. Model O-D: Model O-C + days with WBGT ≥28°C (each model cumulative). CI, confidence interval (HC3 robust); VIF, variance inflation factor; AICc, corrected Akaike information criterion. Using the 2020 Census population denominator (Online Resource 1, Table S3) yields a closely comparable Model O-A estimate; inferential statistics differ because this analysis uses HC3 robust standard errors.

Figure Legends

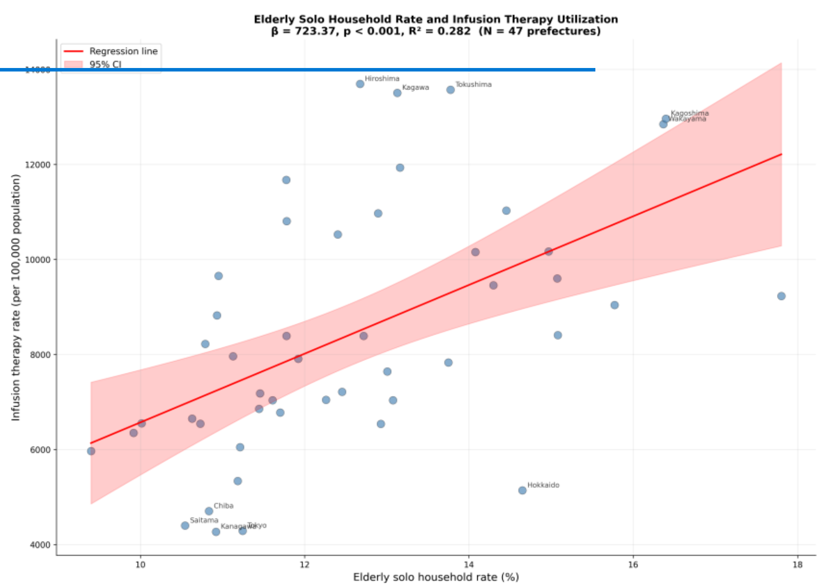
Fig. 1 ~~Scatter~~Coefficient plot ~~showing the association between~~for elderly solo household rate ~~(%) and infusion therapy utilization (per 100,000 population)~~ across ~~47 Japanese prefectures, stratified by the median elderly solo household rate (12.3%). Solid line represents the linear regression fit within each stratum,~~Models O-A through O-D, showing unstandardized coefficients with ~~shaded area indicating~~HC3 95% confidence ~~interval. Top 5 and bottom 5 prefectures by infusion therapy rate are labeled.~~

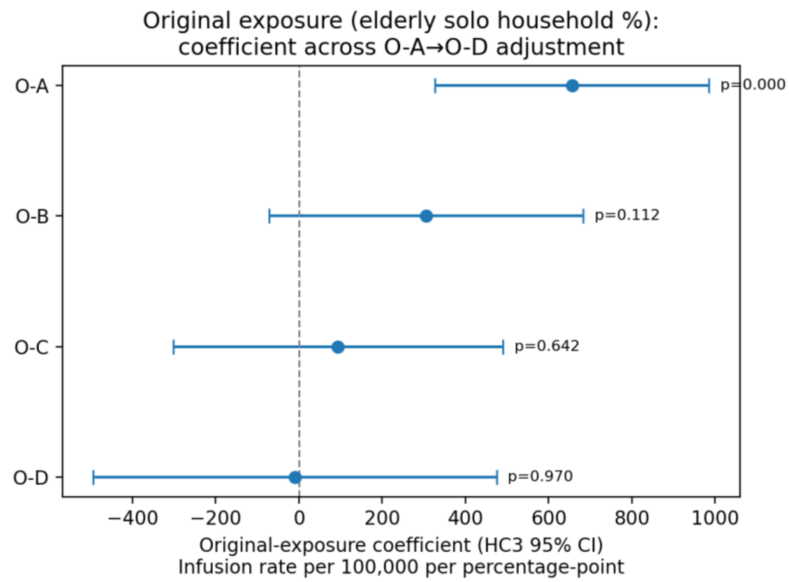
~~**Fig. 2** Scatter plot showing the association between elderly solo household rate (%) and infusion therapy utilization (per 100,000 population) across all 47 Japanese prefectures. Solid line represents the simple linear regression fit (β = 723.37, p = 0.0001, R^2 = 0.282), with shaded area indicating 95% confidence interval~~intervals and a zero reference line.

Fig. 1


Stratified Analysis by Median Elderly Solo Household Rate
High elderly solo HH rate (≥12.3%, N = 24)
β = 264.5, p = 0.458, R² = 0.025
Low elderly solo HH rate (<12.3%, N = 23)
β = 1254.3, p = 0.049, R² = 0.173
Infusion therapy rate (per 100,000)
Elderly solo household rate (%)

~~Fig. 2~~

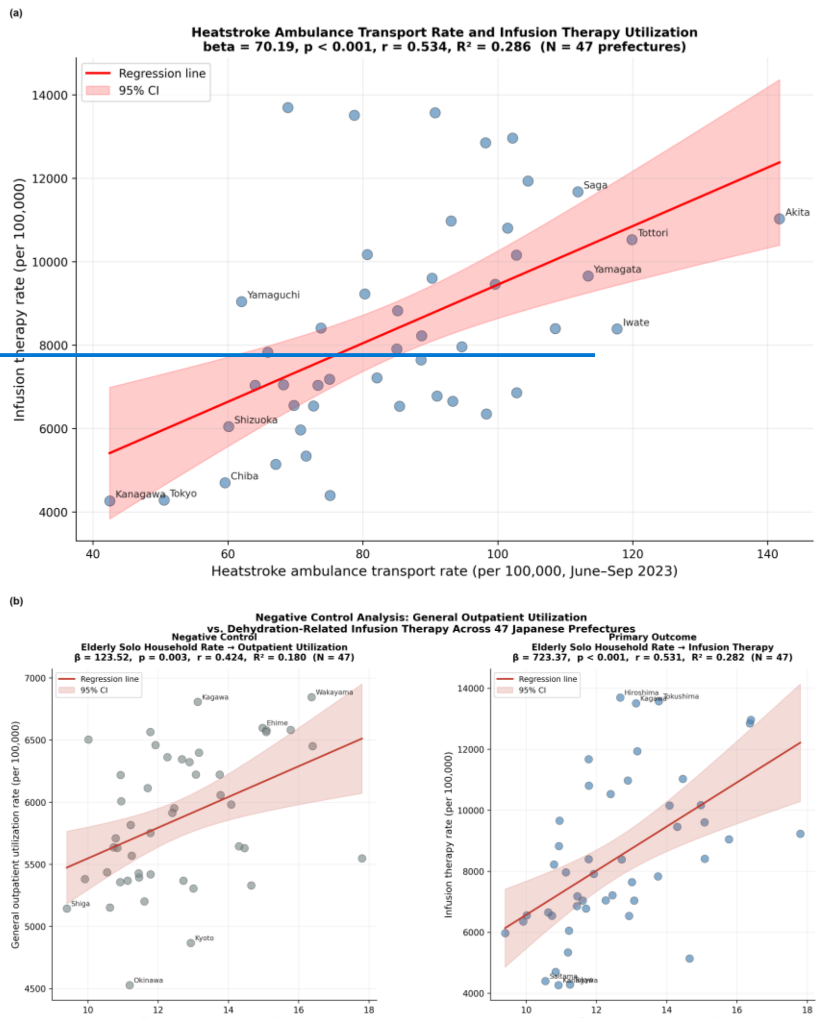

Elderly Solo Household Rate and Infusion Therapy Utilization
β = 723.37, p < 0.001, R² = 0.282 (N = 47 prefectures)
Regression line
95% CI
Hiroshima
Kagawa
Tokushima
Kagoshima
Wakayama
Hokkaido
Chiba
Saitama
Kanagawa
Tokyo
Infusion therapy rate (per 100,000 population)
Elderly solo household rate (%)



Online Resource 1

~~**Online Resource 1** External validation and negative control analyses. (a) Cross-prefectural association between heatstroke ambulance transport rate (Fire and Disaster Management Agency,~~ Online Resource 1 contains supplementary tables and figures, including the 2020 Census-denominator model hierarchy, the alternative-exposure (denominator) analysis, sensitivity and influence diagnostics, nonlinearity analyses (including the median-split analysis), exploratory WBGT metrics with FDR-adjusted p-values, and the comparison-outcome analysis with general outpatient utilization.~~per 100,000 population, June–September 2023) and infusion therapy utilization rate (per 100,000 population; β = 70.19; 95% CI, 36.85–103.53; p < 0.001; r = 0.534), with shaded area indicating 95% confidence interval. Top 5 and bottom 5~~

~~prefectures by heatstroke transport rate are labeled. This cross-validation supports the use of infusion therapy as a population-level indicator of heat-related dehydration. (b) Side-by-side scatter plots comparing the association between elderly solo household rate (%) and general outpatient utilization rate (negative control; R5 Patient Survey; β = 123.52; 95% CI, 44.30–202.74; p = 0.003; R^2 = 0.180; left panel) and infusion therapy utilization rate (primary outcome; NDB Open Data; β = 723.37; 95% CI, 376.52–1,070.21; p < 0.001; R^2 = 0.282; right panel) across 47 Japanese prefectures. The association with infusion therapy was approximately six-fold stronger, supporting partial specificity of the primary association beyond general healthcare access.~~


(a)
Heatstroke Ambulance Transport Rate and Infusion Therapy Utilization
beta = 70.19, p < 0.001, r = 0.534, R² = 0.286 (N = 47 prefectures)
Regression line
95% CI
Infusion therapy rate (per 100,000)
Heatstroke ambulance transport rate (per 100,000, June–Sep 2023)
Saga
Akita
Tottori
Yamagata
Yamaguchi
Iwate
Shizuoka
Chiba
Kanagawa
Tokyo
(b)
Negative Control Analysis: General Outpatient Utilization
vs. Dehydration-Related Infusion Therapy Across 47 Japanese Prefectures
Negative Control
Elderly Solo Household Rate → Outpatient Utilization
β = 123.52, p = 0.003, r = 0.424, R² = 0.180 (N = 47)
General outpatient utilization rate (per 100,000)
Elderly solo household rate (%)
Kagawa
Wakayama
Ehime
Shiga
Kyoto
Okinawa
Primary Outcome
Elderly Solo Household Rate → Infusion Therapy
β = 723.37, p < 0.001, r = 0.531, R² = 0.282 (N = 47)
Infusion therapy rate (per 100,000)
Elderly solo household rate (%)
Hiroshima
Kagawa
Tokushima